National Forensics Sciences University, Goa Campus
TA1 Examination

Program – M.Sc. Cyber Security/Digital forensic and information security
Sem – III
Date - 10-09-2024
Subject Name – Critical Infrastructure security
Subject Code – CTMSCS SIII P4 EL1/ CTMSDFIS SIII P4 EL1
Time- 11:00 A.M. to 11:45 A.M.
Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Multiple Choice Questions (1 mark each)		10 marks
	i.	DCS stands for: a. Distributed central system b. Design control system c. Distributed control systems d. None	1 mark
	ii.	RTU is a: a. Master control unit. b. Human machine interface c. None	1 mark
	iii.	One of PLC functionalities is: a. managing the sensors and actuators b. displays the data to the user c. None	1 mark
	iv.	Data historian in SCADA: a. is a human being. b. is a software. c. Both d. None	1 mark
	v.	The risks involved in OT systems is/are: a. Human Safety. b. Environmental Safety. c. Material damage d. all	1 mark
	vi.	Sensors are the part of which level of PURDUE model: a. Level 0 b. Level 1 c. Level 2 d. None	1 mark
	vii.	Data historian is a part of which level of PURDUE model? a. Level 0 b. Level 1	1 mark

	c. Level 2 d. Level 3	
	viii. How many levels in the PURDUE model: a. 4 b. 5 c. 6 d. 7	1 mark
	ix. ICS-DMZ is the layer for sharing information: a. Between process control zone and operation zone b. Business zone and operation zone c. IT and OT d. None	1 mark
	x. IACS in the context of OT stands for: a. Industrial automation and control system b. Industrial automated centre for security c. International authority of cyber security d. None	1 mark
Q.2	Answer any 3 questions (3x5 marks each)	15 Marks
	i. Explain the difference between OT vs IT?	5 marks
	ii. Explain the two real time case study of OT failure and the reasons behind them.	5 marks
	iii. Explain SCADA architecture with the help of diagram.	5 marks
	iv. What is PURDUE model? Explain with the help of diagram.	5 marks

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme –M.Sc. Cyber Security/ M.Sc. DFIS Sem – III Date- 10/10/2024
Subject Name: Critical Infrastructure Security Subject Code-CTMSCS SIII P4 EL1/ CTMSDFIS SIII P4 EL1
Duration- 1.5 Hours Max.Marks- 50
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Answer any four questions.	20 marks
	a. What are the four core differences between IT and OT?	5 marks
	b. What is OT vulnerability? What are the reasons of OT vulnerability?	5 marks
	c. Why is penetration testing required in OT system? What are the steps involved in it?	5 marks
	d. How the SCADA Protocols evolved?	5 marks
	e. Explain RTU, PLC and IED in the context of SCADA.	5 marks
Q.2	Attempt all	15 marks
	a. Explain the working of MODBUS protocol	5 marks
	b. What is SCADA? Explain its architecture.	5 marks
	c. What are five best practices for OT security?	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Discuss the DNP3 protocol with the help of a diagram	8 marks
	OR	
	II. Discuss any case study related to OT Security failure in detail.	8 marks
Q.3 a	Attempt any one	7 marks
Q3 b	I. What is the importance of CIS in health care sector? Explain with example.	7 marks
	OR	
	II. Explain Single firewall DMZ and dual firewall DMZ in SCADA in detail.	7 marks

End of Paper

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)
M. Sc. Cyber Security, Semester - III

Subject Code: CTMSCS SIII P4 EL3 **Date: 09/12/2024**
Subject Name: CRITICAL INFRASTRUCTURE SECURITY
Time: 10:30 AM – 01:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Attempt any three.**	
(a)	What is SCADA? Highlight key technological advancements that contributed to their growth.	08
(b)	With the help of a diagram, explain the Purdue Model of SCADA system architecture. How do its different layers ensure efficient control and data flow in industrial systems?	08
(c)	Identify and explain at least four common threats or attacks that target ICS/SCADA systems.	08
(d)	Discuss the primary challenges in ensuring the security of ICS/SCADA systems. What strategies can be employed to address these challenges effectively?	08
Q.2	**Attempt any three.**	
(a)	Explain the evolution of SCADA communication protocols and their role in improving system interoperability and reliability.	08
(b)	Explain DNP3 communication architecture by stating its inputs and outputs used.	08
(c)	Compare and contrast the Modbus, DNP3 protocols in terms of functionality, applications, and communication mechanisms.	08
(d)	Discuss the fundamental concepts of PLC programming and explain how ladder logic is used to implement basic industrial control tasks.	08
Q.3	**Attempt any three.**	
(a)	Compare IT security and OT security, emphasizing their differences in priorities, challenges, and implementation strategies.	08
(b)	Briefly explain your team on Stage 1 of the ICS Risk Assessment Framework/Process Steps by taking suitable example wherever necessary.	08
(c)	Identify and explain common vulnerabilities in ICS/SCADA systems and their potential impact on industrial operations.	08
(d)	Describe at least three tools or technologies used in ICS/SCADA penetration testing and their specific roles in ensuring system security.	08
Q.4	**Attempt any two.**	
(a)	Explain the concept of logic injection in ICS/SCADA systems and its potential impact on system functionality and security.	07
(b)	What is asset enumeration in ICS/SCADA systems, and why is it crucial for maintaining cybersecurity and operational reliability?	07

(c) What is Cross Site Scripting & how can an individual perform the same on a SCADA ecosystem? Also, state the threats associated of Cross Site Scripting on SCADA Systems. 07

Q.5 **Attempt any two.**

(a) Discuss the application of the CIA Triad in ensuring the security of ICS/SCADA systems and highlight the challenges in balancing these principles. 07

(b) Write a brief note on ICS – Cyber Kill Chain. 07

(c) Describe the steps involved in integrating risk assessment and scenarios, performing cost-benefit analysis, and establishing a risk mitigation strategy for ICS systems. 07

--- End of Paper---

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. Cyber Security Sem – III Date- 09-09-2024

Subject Name- Blockchain & Cryptocurrencies (CTMSCS SIII P1) Max. Marks- 25

Time- 45 mins

Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1 Multiple Choice Questions (1 mark each) **10 marks**

1. What is the main feature of a Blockchain that ensures data immutability?
a) Centralized control
b) Hashing and cryptography
c) Symmetric encryption
d) Permissionless network

2. Which of the following best describes a Zero-Knowledge Proof (ZKP)?
a) A protocol where the prover can convince the verifier of a statement without revealing the underlying data
b) A method of public-key encryption
c) A technique to share cryptographic keys securely
d) A system for authenticating users through passwords

3. In a blockchain, what is the primary role of consensus mechanisms like Proof of Work (PoW)?
a) To generate random transactions
b) To provide incentives to miners
c) To verify that all participants agree on the state of the blockchain
d) To store public keys of users

4. Which of the following is a characteristic of a Memory Hard Algorithm?
a) It requires high computational power
b) It is difficult to parallelize efficiently
c) It can only run on GPUs
d) It relies solely on disk space for security

5. In blockchain networks, what does the term 'fork' typically refer to?
a) A type of encryption
b) A change in the protocol or software that splits the blockchain
c) A mining strategy to increase rewards
d) The use of multiple private keys for one transaction

6. Which of the following is NOT an advantage of using Zero-Knowledge Proofs (ZKP) in blockchain technology?
a) Reduced computational complexity
b) Increased privacy and anonymity
c) Improved scalability of blockchain

d) The need for a trusted third party

7. Why are memory-hard algorithms considered suitable for applications like password hashing?

a) They use less memory than traditional hashing algorithms
b) They reduce the effectiveness of brute-force attacks by increasing memory requirements
c) They execute faster on modern hardware
d) They are more efficient for lightweight devices like IoT

8. What is the primary difference between public and private blockchain networks?

a) Public blockchains are controlled by a central authority
b) Private blockchains allow only selected participants to validate transactions
c) Public blockchains are permissioned, while private blockchains are permissionless
d) Private blockchains are open to anyone, while public blockchains are restricted

9. What is the primary function of a smart contract in a blockchain network?

a) To manage the hashing process
b) To allow the automatic execution of agreements once certain conditions are met
c) To control access to the network
d) To serve as a wallet for digital currency

10. What is the purpose of sharding in a blockchain network?

a) To reduce transaction fees by compressing blocks using Merkle trees
b) To improve network security by increasing the number of validators for each block
c) To increase the scalability by partitioning the blockchain into smaller pieces
d) To ensure that only full nodes can participate in consensus, thereby enhancing decentralization

Q.2 **Answer any 3 questions (3x5 marks each)** **15 Marks**

A. List the properties of hash function

B. Explain the Byzantine Generals Problem in distributed computing. How does blockchain technology address this problem using consensus mechanisms?

C. Compare the various block chain types.

D. In the class, I had demonstrated a topic by using two coloured toy sets by asking one of the students to volunteer to be on stage. It was about which topic? Explain the topic in relation to the blockchain.

END

Enrolment No.

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme – M. Sc. Cyber Security,
Subject Name: Blockchain and Cryptocurrency (CTMSCS SIII P1) **Sem – III** **Date – 09-10-2024**
Time- 1.5 Hours **Max. Marks- 50**
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1 Solve any four **20 marks**

a. What are the main objectives of cryptography explain the types. 5 marks
b. Directed Acyclic Graph (DAG) in Blockchain Technology 5 marks
c. What are points that need to be consider while designing the Blockchain Network. 5 marks
d. Explain any two types of Consensus mechanisms in blockchain. 5 marks
e. How the transaction fees in Bitcoin and Ethereum are calculated. 5 marks

Q.2 Attempt all **15 marks**

a. Define a memory-hard algorithm and explain its significance in blockchain and cryptographic systems. 5 marks
b. Explain the role of Patricia Trees in blockchain technology. 5 marks
c. Compare the Soft forks and hard forks. 5 marks

Q. 3 a. Explain the concept of a hash pointer in the context of a blockchain. How does it differ from a regular pointer, and what role does it play in ensuring the security and integrity of blockchain data **8 marks**

OR

b. Explain the Key generation and signing process of Elliptic Curve Digital Signature Algorithm (ECDSA). 8 marks

Q.4 a. What are the Key Concepts in Mining of Blockchain. Explain. **7 marks**

OR

b. What is Blockchain governance and Off-chain and On-chain governance. Write a note on Anonymity in blockchain. (4+3) marks

*** All the best***

Seat No.: [redacted]

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)

M. Sc. Cyber Security/Digital Forensic and Information Security

Semester - III

Subject Code: CTMSDFIS SIII P3/CTMSCS SIII P1 **Date: 06/12/2024**

Subject Name: Blockchain and Cryptocurrency

Time: 10:30 AM – 01:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Attempt any three.**	
(a)	What is Quantum Computing and its impact on existing cryptographic methods.	08
(b)	What is Byzantine General Problem and Zero Knowledge Proof.	08
(c)	What are Hash Pointers in block chain. Show the structure of a Hash Pointer.	08
(d)	Explain the concept of a hash pointer in the context of a blockchain. How does it differ from a regular pointer, and what role does it play in ensuring the security and integrity of blockchain data	08
Q.2	**Attempt any three.**	
(a)	Explain an application of the blockchain along with the properties of blockchain.	08
(b)	Write a difference public, private and hybrid blockchains.	08
(c)	Write a note on Ledger, Smart Contracts, Consensus algorithm and Transaction Fee.	08
(d)	Explain any three concepts in blockchain mining.	08
Q.3	**Attempt any three.**	
(a)	Compare the Soft forks and Hard forks in terms of mechanism, examples, advantages and limitations.	08
(b)	What is Nakamoto Consensus and the core principles.	08
(c)	What is the process of Proof of Stake, advantages and limitations.	08
(d)	Write a note on Energy Consumption in Proof of Work. What are the alternatives.	08
Q.4	**Attempt any two.**	
(a)	What is the GHOST Protocol, and why is it necessary? What is a DAO, and how does it function?	07
(b)	Explain the following terms: Gas Limit, and Gas consumption and Gas fees in connection to smart contract.	07
(c)	Write any four possible attacks on blockchain vulnerabilities.	07
Q.5	**Attempt any two.**	
(a)	What is the role of blockchain black market and its impacts in global economy.	07
(b)	Write any one application of blockchain in detail.	07
(c)	Write short note on Difficulty level, 51% vulnerability.	07

--- End of Paper---

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – MSC CS/MSC DFIS | **Sem – 3** | **Date – 11/09/2024**

Subject Name - Social Network Analysis | **Subject Code- CTMSCS SIII P5/CTMSDFIS SIII P5 PE-2**

Time- 45 minutes | **Max. Marks- 25**

Instructions - 1) Answer all questions.

Q.1 **Short Questions (5 mark each)** **10 Marks**

A. Why is OSINT Needed? 5 marks

OR

A. What is Gray literature? 5 marks

B. Discuss OSINT Entities and Organisations. 5 marks

Q.2 **Answer any 3 questions** **15 Marks**

A. Explain Four categories of open information and intelligence as per NATO literature. 8 marks

B. What is OSINT, explain in detail. 7 marks

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Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY, GOA CAMPUS

Social Network Analysis – MSC CS/ DFIS
Mid Semester Sem – 3 Oct-2024

Subject Code: CTMSCS SIII P5/CTMSDFIS SIII P5 PE-2 **Date: 14.10.2024**
Subject Name: - SOCIAL NETWORK ANALYSIS
Time: 11:00 AM TO 12:30 PM

Total Marks: 50

30

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1.	**Answer the following / Define the following** A. Explain about OSINT Entities and Organizations. 6 B. What parties are interested in OSINT information? What advantages are they taking? 6 **OR** C. What are the Challenges of Open-Source Intelligence?	10*2 (Marks)=**20** 12
Q.2.	**Answer the following question in short (any 4)** A. Explain about Information Gathering Types. B. Diseuss benefits of OSINT. C. What are the legal and ethical constraints of OSINT? D. What are Geospatial Information Systems? Explain in brief. E. Explain about Geolocation Tracking, application, advantages.	5*4 (Marks)=**20** 12
Q.3.	1. What constitutes sensitive personal information? A) Publicly available data B) Medical records and financial information C) Job titles D) Social media posts 2. Which of the following is a legal constraint related to the use of OSINT? A) Freedom of Information Act B) GDPR (General Data Protection Regulation) C) Open Records Laws D) All of the above 3. What is a major challenge associated with OSINT? A) Cost of data collection B) Information overload and accuracy C) Lack of technology D) Limited access to public data	**1*10 (Marks)=10** 6

4. Which of the following is a benefit of OSINT?
   A) Low cost compared to classified intelligence
   B) Exclusivity of information
   C) Guaranteed accuracy of data
   D) Minimal legal constraints

5. Which type of OSINT involves monitoring social media platforms?
   A) Technical intelligence
   B) Geospatial intelligence
   C) Human intelligence
   D) Social media intelligence

6. Who are typical parties interested in OSINT information?
   A) Law enforcement agencies
   B) Corporations for market research
   C) Journalists and researchers
   D) All of the above

7. Which type of OSINT is focused on geographical and spatial data?
   A) Cyber intelligence
   B) Geospatial intelligence
   C) Financial intelligence
   D) Human intelligence

8. What ethical consideration should be taken into account when gathering OSINT?
   A) Accuracy of the information
   B) Respecting privacy and consent
   C) The motive behind the data collection
   D) All of the above

9. Which of the following is NOT a type of OSINT?
   A) Technical intelligence
   B) Financial intelligence
   C) Classified intelligence
   D) Media intelligence

10. What is a key challenge in verifying OSINT data?
   A) The abundance of information
   B) Lack of technology tools
   C) Over-reliance on paid sources
   D) Insufficient training

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.Sc. Cyber Security - Semester – III
Semester End Examination – December, 2024

Subject Code: CTMSCS SIII P5 EL1
Subject Name: Social Network Analysis **Date: 10/12/2024**
Time: 10:30 AM – 01:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

No.	Part	Question	Marks
Q.1	(a)	Discuss how to detect and verify fake news with relevant examples.	05
	(b)	Write a note on metadata with an appropriate example related to multimedia and documents.	05
	(c)	Explain various crimes associated with social media and social networks.	07
Q.2	(a)	Explain Google Dorking, including various queries and its implications in online searches.	05
	(b)	Analyze a case study on investigating a fake social media profile.	05
	(c)	Explain different privacy and security settings across various social networks and media platforms.	07
Q.3	(a)	Describe a case where Social Media Forensics or OSINT played a critical role in solving a real-world investigation?	08
	(b)	Discuss open source information categories.	08
		OR	
	(b)	Explain the following terms: Web Archive, API, and Nodes and Edges in Graph Theory.	08
Q.4	(a)	Discuss the challenges of Open-Source Intelligence (OSINT).	05
	(b)	Discuss the difference between social media and social network.	05
	(c)	Write a note on Darkweb investigation.	07
		OR	
	(c)	Explain Username, Email ID, and Mobile number OSINT with examples.	07

Q.5	(a)	What kind of information can be investigated using Username, IP and email ID OSINT?	05	
	(b)	Explain the ethical and legal constraints of OSINT.	05	
	(c)	Explain any 7 impacts of social media on Business.	07	
Q.6	(a)	Explain various types of information gathering in OSINT with relevant examples.	08	
	(b)	Write a case study on social media fake profile investigation.	08	
		OR		
	(b)	Explain Four categories of open information and intelligence as per NATO literature.	08	

******End of Paper******

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. CS	Sem – III	Date- 10.09.24
Subject Name- Cloud Security & Forensics	Subject Code- CTMSCS S3-P3	
Time- 45 minutes		Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	Multiple Choice Questions (1 mark each)	10 marks
	1a) While indexing the data content to enhance the search performance we can avoid the inclusion of sensitive data by using: a. Low protection b. Medium protection c. Strong Protection **d.** None of these	1 mark
	1b) Among the following, which attack can be performed at the application layer? a. Rootkit b. Buffer overflow c. Trojans d. TOCTTOU	1 mark
	1c) Which of the following provides the least security **a.** IaaS **b.** PaaS **c.** SaaS **d.** None	1 mark
	1d) The risk associated with Infrastructure as a Service can be minimised by using: a. PALM b. HyperSafe c. Closed box execution environment **d.** Access control	1 mark
	1e) Policy ranking is the: **a.** Ranking of different policies **b.** Indexing of data **c.** Tool to help the user to select CSP **d.** None	1 mark
	1f) In **cloud computing JAR files are used to:** **a.** To protect data **b.** To execute the data **c.** To avoid virtualization threat **d.** None	1 mark
	1g) What is a possible risk of cloud computing **a.** Lack of access to data **b.** Storage of data without control over the location of where the data is stored **c.** Lack of ability to back up data d. None	1 mark

	1h) CSP need not to decrypt the data while performing the search operation by using: **a.** ECC **b.** RSA **c.** Searchable encryption technique **d.** All are correct	1 mark
	1i) Among the following which not a public cloud: **a.** AWS **b.** Azure **c.** nextCloud **d.** All are correct	1 mark
	1j) Among the following which hypervisor is a bare metal hypervisor. a. Oracle VB b. KVM c. Vmware workstation d. None of these	1 mark
Q.2	Answer any 3 questions (3x5 marks each)	15 Marks
	i. What is multi cloud explain with suitable example and also discuss the security risk associated with it?	5 marks
	ii. Discuss and draw the NIST cloud security architecture model.	5 marks
	iii. What is indexing in cloud environment? Why do we need it, discuss in detail?	5 marks
	iv. Discuss the possible vulnerabilities with respect to layers in cloud environment.	5 marks

National Forensics Sciences University, Goa Campus
Mid-semester Examination

Program – M. Sc. Cyber Security Sem – III Date- 07.10.24
Subject Name- Cloud Security & Forensics Subject Code- CTMSCS S3-P3
Time- 1.5 Hours Max.Marks- 50
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Solve any four	20 marks
	a. Explain block-based storage in cloud computing.	5 marks
	b. With a neat diagram explain each component of the Xen hypervisor.	5 marks
	c. Explain the concept of virtualization layers.	5 marks
	d. Explain the different components of SOC-2 certification.	5 marks
	e. How hardening helps to make our system secure. Explain by considering the use case of software application hardening.	5 marks
Q.2	Attempt all	15 marks
	a. How object-based storage is different from file-based storage.	5 marks
	b. How do you achieve operating system hardening, explain.	5 marks
	c. What is Para virtualization? List out at least three hypervisors based on it.	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Differentiate between KVM, VMware ESXi, Hyper V, and Xen hypervisor.	8 marks
	OR	
	II. Explain PCI passthrough in detail.	8 marks
Q.3 b	Attempt any one	7 marks
Q3 b	I. How do you investigate a case on Amazon Cloud Drive?	7 marks
	OR	
	II. What is a Docker? How containers are different from images, explain with a suitable example.	7 marks

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security - Semester - III - DEC-2024

Subject Code: CTMSCS SIII P3 **Date: 04/12/2024**

Subject Name: Cloud Security and Forensics

Time:10.30 AM To1.30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Attempt any three.**	
(a)	Compare Discretionary Access Control and Mandatory Access Control Mechanism? What is the limitation of RBAC ? How does ABAC overcome the limitation of RBAC?	08
(b)	How Docker is different from the virtual machine? Discuss different components of Docker Environment.	08
(c)	List out and briefly discuss the functions of following Cloud Infrastructure a. Hypervisor b. Cloud Broker	08
(d)	Compare the working of SaaS and PaaS based on following parameters : Services, Provider, Users, Limitation, Security, billing	08
Q.2	**Attempt any three.**	
(a)	If you were to switch to cloud computing what considerations might cross your mind? List out and Briefly discuss possible strategies in cloud migration.?	08
(b)	Answer the following questions briefly : (Any TWO) a. Discuss the challenges of Cloud Forensics. b. Which are the different sources of artifacts in Cloud Eco-system.? C Which are the different sources of Logs in Cloud Computing Environment?	08
(c)	List out and discuss cloud computing Vulnerabilities. What are the possible prevention techniques for cloud threat?	08
(d)	Compare the following terms (Atleast Four points for each would be required): a. Symmetric vs Asymmetric Cipher b. Stream vs. Block Cipher	08
Q.3	**Attempt any three.**	
(a)	Which services are provided by the IaaS provider? How Bare Metal as a	08

Service differs from Infrastructure as a Service? Discuss advantages and disadvantages of IaaS service model

(b) Suppose the prime number $p = 17$ and a primitive root of p is g=3. If Alice chooses her private key $x = 5$, and Bob chooses his private key 7, what is the shared secret key? 08

(c) Discuss different memory management techniques to optimize virtual memory of Hypervisor. 08

(d) What is virtualization? Discuss Full Virtualization in detail with its Implementation techniques 08

Q.4 **Attempt All.**

(a) Compare cloud deployment models based on their advantages and disadvantages : Public cloud, Private cloud and community cloud 07

OR

Discuss cloud forensic architecture in detail.

(b) What is SLA? Discuss different types of SLA. Which are the essential components of a cloud SLA? 07

OR

Explain the working of Google App Engine architecture.

Q.5 **Attempt All.**

(a) Which characteristics of Cloud computing make it different from traditional data centers? Discuss different ways to achieve scalability in cloud environment. 07

(b) P and Q are TWO prime numbers. P=7, and Q=19. Provide confidentiality using RSA, for encrypting & decrypting message (M= 'F'). Please consider possible value set for e= {15, 6, 48, 5, 81} and d= {89, 77, 47,65}. 07

END OF PAPER

National Forensics Sciences University, Goa Campus
TA1 Examination

Program – M.Sc. Cyber Security/Digital forensic and information security
Sem – III
Date - 09-09-2023
Subject Name – IoT Security and Forensics
Subject Code – CTMSCS SIII P2/ CTMSDFIS SIII P2
Time- 03:00 P.M. to 03:45 P.M.
Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Multiple Choice Questions (1 mark each)		10 marks
	i.	"Internet of things" term first coined in: a. 1998 b. 1999 c. 2000 d. 2001	1 mark
	ii.	Who coined the term "Internet of Things"? a. Kevin Aston b. John wright c. Edward jameson d. None	1 mark
	iii.	Which of the following is not an application of IoT a. sensors b. self-driven car c. smart city d. smart home	1 mark
	iv.	Which layer is used for wireless connection in IoT devices? a. Application layer b. Network layer c. Data link layer d. None	1 mark
	v.	Which of the following is used to capture data from the physical world in IoT devices? a. Sensors b. Actuators c. Microcontrollers d. all	1 mark
	vi.	Which of the following protocol is used to link all the devices in the IoT? a. HTTP b. UDP c. Network	1 mark

		d. TCP/IP	
	vii.	IoT systems can be categorized in: a. 4 Levels b. 5 Levels c. 6 Levels d. 7 Levels	1 mark
	viii.	Cloud storage is not used in which level of IoT systems a. Level 2 b. Level 3 c. Level 4 d. None	1 mark
	ix.	Which of the following is not a fundamental component of an IoT system? a. Sensors b. Connectivity and data processing c. User interface d. Transformers	1 mark
	x.	Which of the following is false about IoT devices? a. IoT devices use the internet for collecting and sharing data b. IoT devices need microcontrollers c. IoT devices use wireless technology d. IoT devices are completely safe	1 mark
Q.2	Answer any 3 questions (3x5 marks each)		15 Marks
	i.	Discuss the evolution of IoT?	5 marks
	ii.	Explain about the major component IoT.	5 marks
	iii.	What are the important features of IoT.	5 Marks
	iv.	Explain the different levels of IoT deployment.	5 marks

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme – M.Sc. Cyber Security / M.Sc. DFIS
Sem – III **Date- 08/10/2024**
Subject Name: IoT Security and Forensics **Subject Code- CTMSCS SIII P2/ CTMSDFIS SIII P2**
Duration- 1.5 Hours **Max. Marks- 50**
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Answer any four questions.	20 marks
	a. What are the challenges in implementation of IoT?	5 marks
	b. What is QoS requirement in M2M?	5 marks
	c. Which technologies could be helpful in implementation of IoT system. Briefly explain.	5 marks
	d. There are different types of nodes in wireless sensor network. Which types of nodes are favourable and which are not and why?	5 marks
	e. Differentiate between IoT and M2M.	5 marks
Q.2	Attempt all	15 marks
	a. Which are the different components of IoT? Explain it with respect to any one IoT application.	5 marks
	b. Draw IoT protocol stack and explain each layer in brief.	5 marks
	c. What are the important characteristics of M2M. Explain with example.	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Differentiate between Zwave and Zigbee.	8 marks
	OR	
	II. Explain features and advantages of 6LoWPAN.	8 marks
Q.3 a	Attempt any one	7 marks
Q3 b	I. Explain Bluetooth Protocol Stack. How it is different from IoT Protocol Stack	7 marks
	OR	
	II. Explain why IoT is crucial from a nation security point of view.	7 marks

End of Paper

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)
M.Sc. Cyber Security
Semester – III

Date: 05/12/2024

Subject Code: CTMSCS SIII P2
Subject Name: IoT Security and Forensics
Time: 10.30AM to 1.30PM
Total Marks: 100

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Attempt any three.**	
(a)	Draw IoT protocol stack. Write about function and components of each stack.	**08**
(b)	Write about penetration testing of IoT devices in detail.	**08**
(c)	Which are the different phases of IoT Forensics? Explain any four phases in detail.	**08**
(d)	What is the goal of threat modelling in IoT? What are the different phases of IoT threat modelling? Explain each phase in detail.	**08**
Q.2	**Attempt any three.**	
(a)	What is logical design IoT based system? What are different kinds of logical communication models? Explain any two models in detail.	**08**
(b)	What are the major components of MQTT protocol? Explain functions of each components.	**08**
(c)	How SDN can be helpful as compared to conventional networking scheme? Explain in detail.	**08**
(d)	Write about challenges in IoT security.	**08**
Q.3	**Attempt any three.**	
(a)	Write features of any one controller used for IoT based application. Write Arduino functions used for Input & Output of analog & digital signals with suitable example.	**08**
(b)	Write a note on SNMP.	**08**
(c)	Explain about CoAP communication protocol in detail.	**08**
(d)	Write features of XMPP and explain architecture of XMPP protocol.	**08**
Q.4	**Attempt any two.**	
(a)	What are the vulnerabilities mentioned in Open Web Application Security Project? Explain top five vulnerabilities and its mitigation techniques in detail.	**07**

(b) Draw and explain SDN architecture and function of each block in detail. 07

(c) Which are the different attack surface in IoT system? Explain each attack surface along with its attack vector. 07

Q.5 **Attempt any two.**

(a) Write about HTTP communication protocol with functions used for data communication. 07

(b) What are the different levels of IoT forensics? What kinds of artifacts can be extracted from each level? 07

(c) Write features of any four IoT forensic tools and its limitations. 07

END OF PAPER